

2026-06-03

Social Gaze Task

이전 Meeting 내용 및 Action item

<Social Gaze>

- Gazelle VSGaze dataset에 별도 finetune해 나온 ap 등 성능 공유, 재현한 MTGS 성능 공유
- Task-specialized graph 구조 생각한 것, VLM/LLM graph prompting을 통한 candidate disambiguation 아이디어 공유

Action item

<Social Gaze>

- 진웅님 Graph Implementation 방식 이해, social gaze에 더 적합한 방향으로 고민.
- 해당 graph 방식과 vlm/llm 연결해 일단 vlm/llm 최종 prediction하는 방향 생각해보기

Beyond Prefixes: Graph-as-Memory Cross-Attention for Knowledge Graph Completion with Large Language Models

Ruitong Liu¹, Boxu Lin², Peize Li³, Siyuan Li^{4,*}, Yunjia Wu², Te Sun⁵, Chaohan Wu²

¹ Peking University ² Dalian University of Technology ³ King's College London

⁴ Peng Cheng Laboratory ⁵ Shanghai Jiao Tong University

ruitong.jerry@gmail.com, yuanlsy@mail.dlut.edu.cn

* Corresponding author

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핵심 질문: *How to effectively fuse explicit KG structure with implicit LLM semantics at a deep, feature-interactive level?*

기존: Shallow Prefix

[Graph prefix tokens] + [Text prompt] → LLM
그래프 정보를 LLM prompt 앞에 붙이는 방식은 얕다.
Prefix가 약하게 연관되어 implicit reasoning 필요.



제안: Graph-as-Memory

Text prompt → LLM hidden states → query
Graph memory tokens → key/value
↓ **cross-attention**으로 필요한 graph evidence만 읽음
Graph를 외부 메모리로, LLM이 cross-attn으로 읽음.

문제 정의: Knowledge Graph Completion

Knowledge Graph Triple

(h, r, t) — head, relation, tail

예시:

(Barack Obama, born_in, Hawaii)

(Aspirin, treats, Headache)

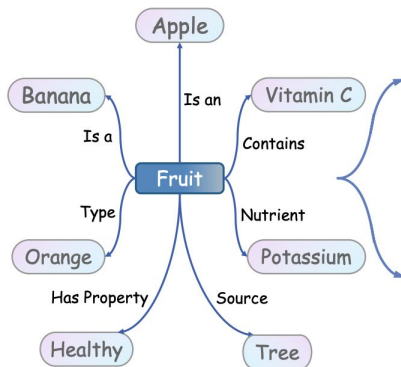
(Fruit, contains, Vitamin C)

Tasks

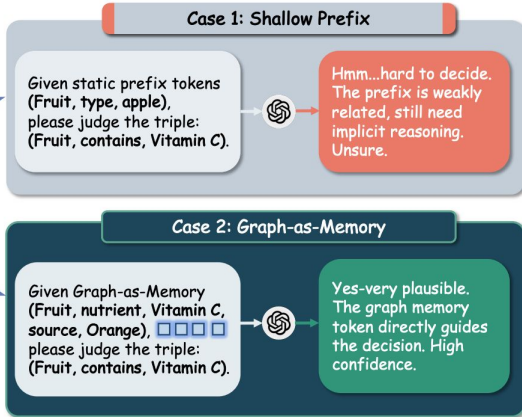
Link Prediction: (h, r, ?) → t 예측

Triple Classification: (h, r, t) → true / false

Knowledge Graph Context



Reasoning Comparison



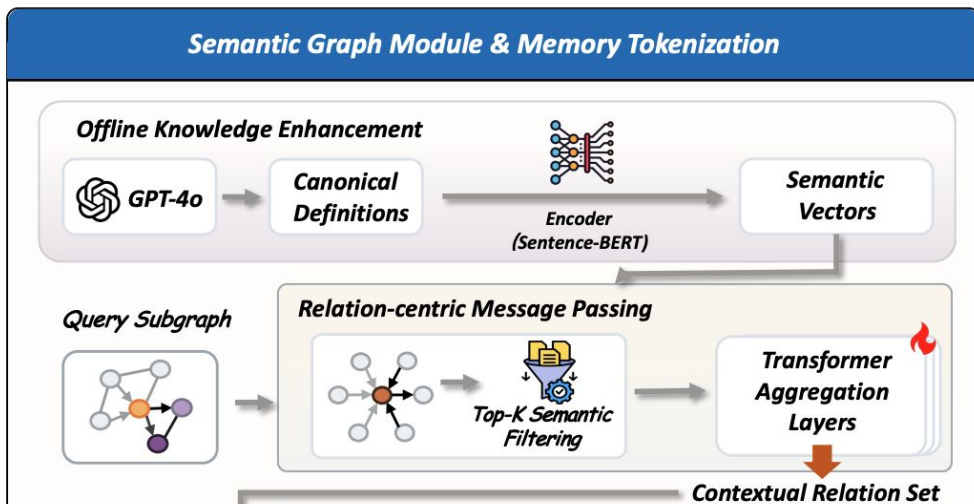
기존 Graph-LLM 연결 방식의 한계

graph embedding → projection → **prefix token** → text prompt와 **concat** → LLM self-attention으로 처리

문제: **relation semantics**가 **local graph context**에 따라 달라지므로, 단순 concat이 아닌 context-dependent graph evidence retrieval이 필요

주요 방법: Semantic Graph Module (SGM)

Graph neighborhood를 LLM이 읽을 수 있는 memory로 변환하는 전단계



Offline Knowledge Enhancement

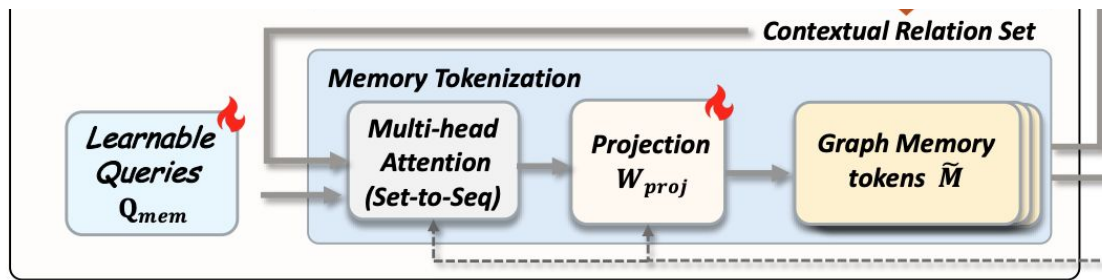
각 **relation type**에 대해 GPT-4o로
canonical **definition** 생성
→ Sentence-BERT로 embedding
→ **Semantic Vectors** 생성

Relation-centric Message Passing

Query triple 기반 local neighborhood
→ Relation semantic vector간 cos sim으로
관련 h/t 주변 edge만 Top-K 선택
→ Transformer-style mean-pool aggregate

주요 방법: Semantic Graph Module (SGM)

Graph neighborhood를 LLM이 읽을 수 있는 memory로 변환하는 전단계

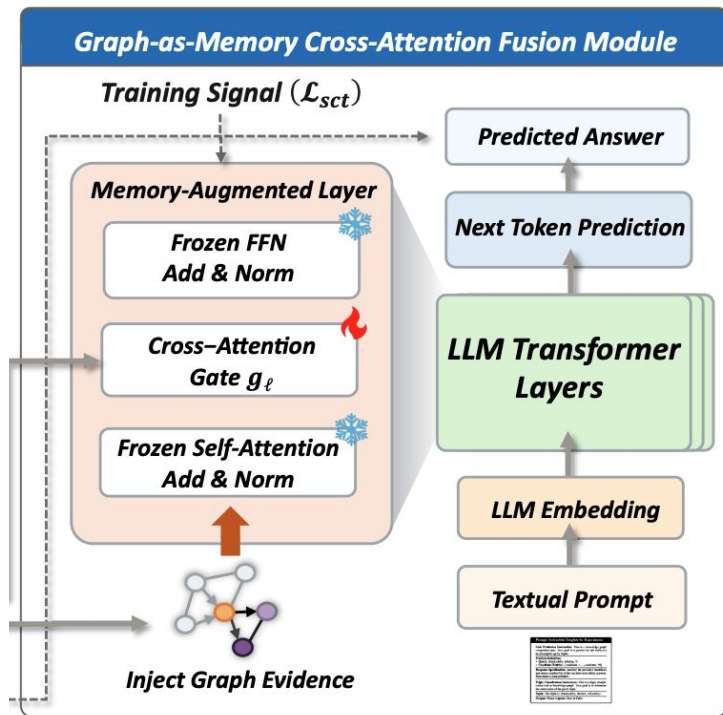


Memory Tokenization

Query마다 relation edge 수가 달라지는 문제 → Learnable query (Q_{mem})로 set-to-sequence compression
→ Multi-head Attention (Set-to-Seq) → Projection W_{proj} → 고정 크기 32개 Graph Memory tokens $\tilde{M}$ 생성
→ LLM hidden size에 맞춤

Stage 1: SGM self-supervised pretraining — positive triple (h,r,t) / corrupted negative triple로 scoring 학습

주요 방법: Graph-as-Memory Tuning



Prompt Templates

Link Prediction: "Predict the tail entity for (head_entity, relation, ?). Candidates: {c1, ..., c20}. Return top 3."

Triple Classification: "The triple is: (head, relation, tail). Please response True or False."

Memory-Augmented Layer 구조

LLM hidden state를 query로,
Graph memory token을 key/value로 cross-attention

Frozen Self-Attention → Add & Norm

→ **Cross-Attention (Gate g_ℓ)** 🔥

→ Add & Norm → Frozen FFN

→ **Next Token Prediction**

Base LLM (Alpaca-7B)은 freeze 상태 유지

Stage 2: GMT Finetune Query subgraph → Graph memory $\tilde{M}$ 생성 → Text prompt LLM 입력 → Memory cross-attn으로 graph evidence 주입 → Answer token generation → Loss: autoregressive next-token prediction

Framework Overview – Social Gaze를 위한 Graph+LLM

1. Visual Evidence Extractor

MTGS 기존 방식 활용해 video clip에서
사람별 gaze evidence 추출:

- P_i의 person token (h_i)
- Gaze direction (g_i)
- Gaze heatmap (H_i)
- In/out score (P_{out})

2. Directed Gaze Graph Builder

Social gaze relation 후보 공간 구축:

- Person nodes: P1, P2, P3, ...
- Region nodes: R1, R2, ..., RK
(grid/gaze heatmap peak clustering)
- Null node: NULL (프레임 바깥)

Edge Construction — 모든 source person에서 모든 target으로 directed edge 생성

source \ target	P1	P2	P3	R1	R2	NULL
P1	—	e_{12}	e_{13}	e_{1r1}	e_{1r2}	e_{1n}
P2	e_{21}	—	e_{23}	e_{2r1}	e_{2r2}	e_{2n}
P3	e_{31}	e_{32}	—	e_{3r1}	e_{3r2}	e_{3n}

Framework Overview

Edge Initialization

각 edge는 두 가지 evidence로 시작: ① heatmap-target overlap (s_{hm}) ② gaze-target alignment (align)

Initial edge state: $e_{i \rightarrow \square}^0 = \text{MLP_init}(\text{source node } P_i, \text{target node } t, \text{edge evidence } [s_{hm}, \text{align}])$

Incoming role: target t 는 누구에게 attention을 받는가

Outgoing role: P_i 가 무엇을 보는가

		target					
		P1	P2	P3	R1	R2	NULL
source							
P1		-	e12	e13	e1R1	e1R2	e1N
P2		e21	-	e23	e2R1	e2R2	e2N
P3		e31	e32	-	e3R1	e3R2	e3N

Framework Overview

3. Dual-Role Edge Refinement

- Initial edge state $e^0_{i \rightarrow t}$ → **row-wise outgoing update** → column-wise incoming update → node update + edge refresh → refined graph evidence $\hat{E} = \{e^L_{i \rightarrow t}\}$

		target					
		P1	P2	P3	R1	R2	NULL
Outgoing role: P_i 가 무엇을 보는가	source						
	P1	-	e12	e13	e1R1	e1R2	e1N
	P2	e21	-	e23	e2R1	e2R2	e2N
	P3	e31	e32	-	e3R1	e3R2	e3N

Outgoing update:
row-wise transformer

같은 row edge들을 하나의 sequence로 만들어 Transformer encoder layer 적용
→ 같은 source에서 나가는 다른 target 후보들끼리 attend.

Framework Overview

3. Dual-Role Edge Refinement

- Initial edge state $e^0_{i \rightarrow t}$ → row-wise outgoing update → **column-wise incoming update** → node update + edge refresh → refined graph evidence $\hat{E} = \{e^L_{i \rightarrow t}\}$

Incoming role: target t 는 누구에게 attention을 받는가

	target					
	P1	P2	P3	R1	R2	NULL
source						
P1	-	e12	e13	e1R1	e1R2	e1N
P2	e21	-	e23	e2R1	e2R2	e2N
P3	e31	e32	-	e3R1	e3R2	e3N

Incoming update:
column-wise transformer

같은 column edge들을 sequence로 만들어 transformer encoder layer 적용
→ 같은 **target**을 향하는 다른 **source** edge들끼리 **attend**

Framework Overview

3. Dual-Role Edge Refinement

- Initial edge state $e^0_{i \rightarrow t}$ $\rightarrow$ row-wise outgoing update $\rightarrow$ column-wise incoming update $\rightarrow$ **node update + edge refresh** $\rightarrow$ refined graph evidence $\hat{E} = \{e^L_{i \rightarrow t}\}$

Source Node Update

자신이 내보낸(outgoing) edges를 모음으로써 어디를 보려는지에 대한 context를 갖게 됨.

: $P_i \leftarrow \text{pool}_t e_{i \rightarrow \square}$

Target Node Update

자신에게 들어온(incoming) edges를 모음으로써 누가 자신을 보는지에 대한 context를 갖게 됨.

: $t \leftarrow \text{pool}_i e_{i \rightarrow \square}$

Edge Refresh: $e^{t+1}_{i \rightarrow \square} = \text{Refresh}(\hat{e}_{i \rightarrow \square}, \text{updated } P_i \text{ node}, \text{updated target } t \text{ node}) \rightarrow \text{반복하며 상호작용}$

Framework Overview

3. Dual-Role Edge Refinement

Source-side target choice ↔ Target-side received attention 반복 상호작용

예: $P_2 \rightarrow P_1$ edge는 updated P_2 의 source context (P_2 가 어디를 보려하는지)와 updated P_1 의 target context (P_1 이 누구에게 주목받는지)를 다시 읽고 더 정확한 relation evidence가 됨

$\hat{E}$ = *socially contextualized directed gaze evidence*

Graph Pretraining

Refined graph evidence $\hat{E}$ 만으로 pretrain: P_i 가 t 를 보는가 1/0 예측하도록 BCE loss

→ Graph 자체가 의미있는 gaze relation evidence를 담도록 학습

Framework Overview

4. Graph Evidence Tokenizer

Variable-size refined edge set $\rightarrow$ Fixed-size LLM evidence tokens

Edge Token 구성

edge token for $P_i \rightarrow t$:

```
[ ei $\rightarrow$ □,           // refined relation vector  
  source_id_emb(i), // source = Pi  
  target_id_emb(t), // target = t  
  target_type_emb(t) // person / region / null  
]
```

예: $P1 \rightarrow P3$ token =

[refined vector, src=P1, tgt=P3, type=person]

Set-to-Token Attention

$S_edge = \{ \text{edge_token}(i,t) \text{ for all } P_i \rightarrow t \}$

$Q_g = \{ q_1, q_2, \dots, q_M \}$ (learnable queries)

$G = \text{Attn}(Q_g, S_edge, S_edge)$

Q = learnable evidence queries

K = refined edge tokens

V = refined edge tokens

$G = \{ g_1, g_2, \dots, g_M \}$

$\rightarrow G_LLM = G \times W_proj$

핵심: edge 수가 sample마다 다르고 많기 때문에 다 넣으면 길고 noisy

$\rightarrow$ Learnable query를 두고 refined edge set에서 중요 graph evidence만 뽑아 fixed-size tokens로 압축

Framework Overview

5. Evidence-Augmented LLM-Reasoner

Input 1: Natural-language Query

LLM hidden states from query가 Q 가 됨

$$\text{Emb}(\langle P_i \rangle) = \text{LLM_token_emb}(\langle P_i \rangle) + W_{\text{node}} \times v_i \text{ (graph person node)}$$

Input 2: Graph Evidence Tokens G_{LLM}

G_{LLM} 이 K/V 가 됨

LLM cross-attention을 통해
query에 맞는 evidence만 retrieve

Query Types → LLM이 같은 G_{LLM} 을 다르게 attend

LAH

Does $\langle P1 \rangle$ look at $\langle P3 \rangle$?

LAEO

Do $\langle P1 \rangle$ and $\langle P3 \rangle$
look at each other?

SA

Do $\langle P1 \rangle$ and $\langle P3 \rangle$
attend to same target?

Training Loss

정답 pos → $L_{\text{LLM}} = -\log P(\text{"Yes"} \mid q, G)$ | 정답 neg → $L_{\text{LLM}} = -\log P(\text{"No"} \mid q, G)$
LLM loss가 내려가려면 G 가 query answer에 도움이 되는 edge evidence를 담아야 함 → learnable queries가
useful edge tokens에 더 잘 attend하도록 학습됨

Framework Overview

Training Stages

Visual-Graph Pretraining →

$L_{\text{heatmap}} + L_{\text{inout}} + L_{\text{graph}}$

- Visual Evidence Extractor
- Directed Gaze Graph Builder
- Dual-Role Edge Refinement

**Graph 구조가 gaze evidence를
잘 읽도록 pretrain**

LLM Alignment →

L_{LLM} (Yes/No next-token loss)

- Graph Evidence Tokenizer
- Evidence-Augmented LLM
- Cross-attention over G

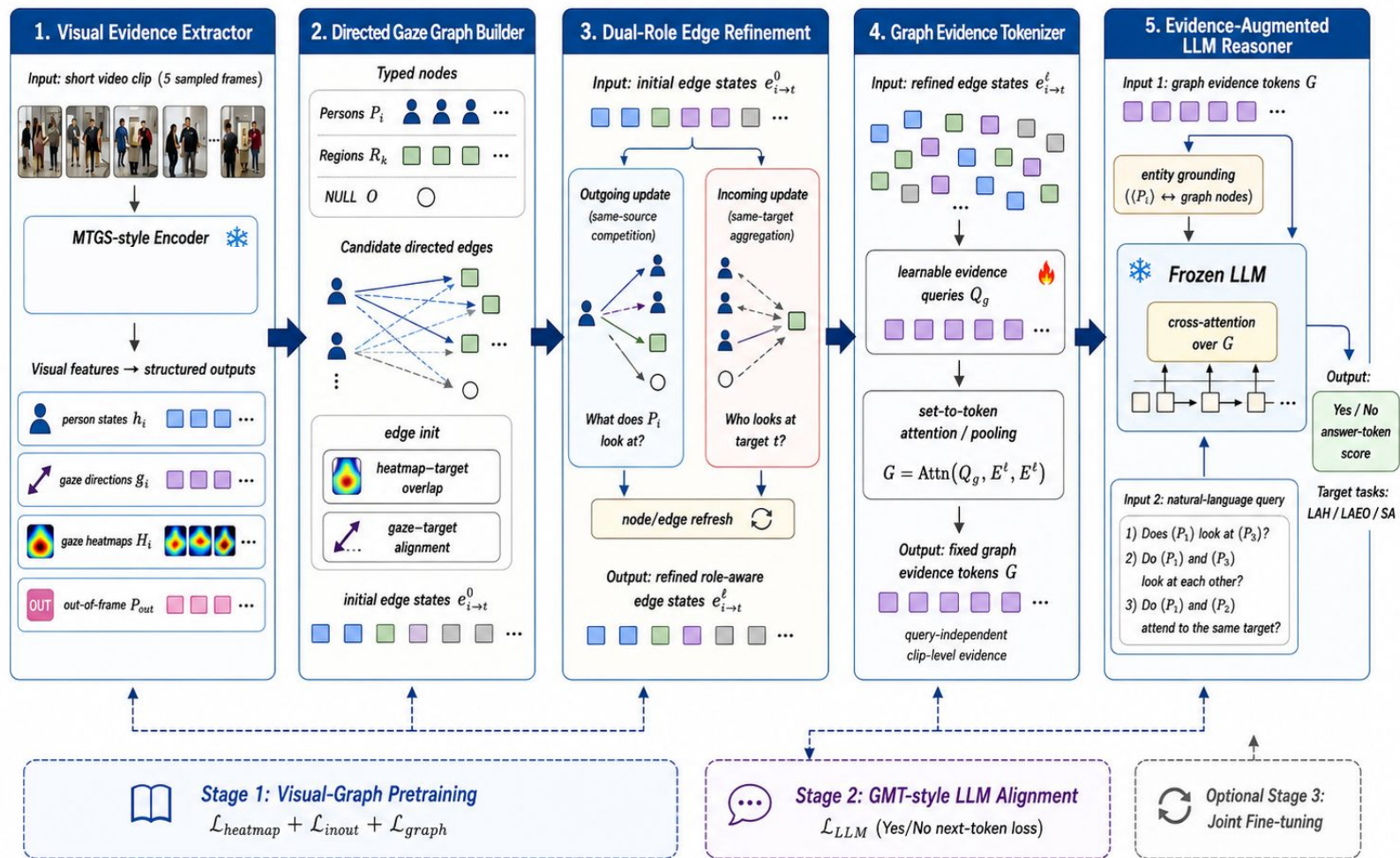
**Graph evidence tokens를
LLM이 읽을 수 있도록 align**

Joint Fine-tuning

전체 파이프라인을

end-to-end fine-tuning

- 모든 모듈 jointly 학습
- **Graph + LLM 상호작용
최적화**



Emotion Task

Framework Overview

Problem

Multimodal emotion recognition task에서 모델은 GT와 유사한 **cue**를 공유하는 **confuser label**로 판단해 틀리게 된다. 모델이 더 **discriminative**한 **cue**에 집중해 **reasoning**해야 모호한 decision boundary가 명확해진다.

Method

- **K-Rollout Generation : Rationale SFT checkpoint에서 sample당 K개의 response를 생성한다.** 각 rollout: <think> rationale </think> <answer> emotion </answer>
- **Sample-Specific Confuser Determination**
 - **Wrong rollout이 존재할 때 : rollout에서 가장 빈번한 wrong label이 confuser**
 - **All correct일 때 : 모든 non-GT label에 대한 π_{ref} margin에서 가장 가까운 label이 confuser**

Framework Overview

Method

- **Confuser-Relative Evidence Score (G)**

1. Observable Cue Parsing

- **Rationale → GPT-4o-mini cue parser → T/A/V cue groups**

T = text/context cues (e.g., unexpected news, argument)

A = audio cues (e.g., sudden rise in pitch, trembling voice)

V = visual cues (e.g., widened eyes, furrowed brow)

Emotion words, interpretation, conclusion → 제외

2. Cue Intervention

- **추출된 T/A/V cue 전체를 frozen π_{ref} 에 넣고 GT-confuser margin을 측정한다.**
- Cues: [unexpected news, sudden pitch rise, widened eyes], GT = surprise, confuser = joy,
 $P(\text{surprise} \mid \text{cues}) = 0.72$, $P(\text{joy} \mid \text{cues}) = 0.15$, $v(\text{TAV}) = \log(0.72) - \log(0.15) = +1.57$
- **양수면 GT쪽, 음수면 confuser 쪽. 클수록 discriminative**
- **$G(R) = v(\text{TAV})$ 사용된 cue가 GT를 confuser로부터 얼마나 분리하는가**

Framework Overview

Method

G-score $\times$ answer correctness로 모든 response를 네 region으로

	Correct Answer	Wrong Answer
High G	Ideal	Mapping Failure
Low G	Fragile	Needs Correction

- **G-Guided Preference Optimization**

- 1. Ideal > Fragile**

Chosen: correct answer + high-G rationale

Rejected: correct answer + low-G rationale

- 3. Ideal > Mapping Failure**

Chosen: correct answer + high-G rationale

Rejected: wrong answer + high-G rationale

- 2. Ideal > Needs Correction**

Chosen: correct answer + high-G rationale

Rejected: wrong answer + low-G rationale

- 4. No Ideal**

Chosen: GT answer + rationale

Rejected: Fragile/Mapp/Needs Corr